

Efficient clustering approach for adaptive unsupervised colour image segmentation

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Abstract: This study proposes a clustering-based colour image segmentation approach consisting of a novel initialisation technique. Colour image segmentation transforms image pixels into regions and a prerequisite for image analysis and computer vision applications. Therefore, colour image segmentation is considered one of the most important processes in image understanding and pattern recognition. This study presents an efficient and adaptive unsupervised approach based on bottom-up red–green–blue (RGB) colour histogram search approach to achieve colour image segmentation. Firstly, the RGB histogram is processed through a double-scan procedure to determine significant modes in each histogram. In the next step, each mode is processed through a bottom-up histogram search approach, completing RGB triplet. The RGB triplets are utilised as the cluster centroids, clustering the pixels into regions and producing the final segmented image. The authors proposed method was compared with several other unsupervised image segmentation algorithms with an extensive experiment performed on various image segmentation evaluation benchmarks. Experimental results show that the proposed algorithm outperforms state-of-the-art algorithms both in terms of features integrity and execution speed.

1 Introduction

Image segmentation is an extremely important preliminary step in a lot of pattern recognition and computer vision applications. Image segmentation divides the image into segments such that each pixel in a segment is homogeneous to all the other pixels contained by that segment [1]. Image segmentation techniques can be classified into threshold-based, region-based and clustering-based approaches [2]. Clustering techniques have a wide application in transforming pixels into regions using the similarity features (e.g. colour, texture etc.). Clustering is an unsupervised approach, having no prerequisite of labelled class data for clustering unknown data. K-means [3] and fuzzy c-means (FCM) [4] are the most popular and widely used clustering techniques for image segmentation.

An important prerequisite for applying clustering techniques to an image is the provision of initialisation parameters, i.e. cluster centroids and the number of clusters. The final segmented image quality heavily depends upon the initialisation parameters. The process of initialisation parameters determination is a challenging task both in terms of preserving the significant image features and also maintaining low computational complexity. Fukunaga and Hostetler [5] presented a non-parametric clustering algorithm with exemption from a pre-defined number of cluster centroids. Mean shift (MS) algorithm has the computational complexity of $O(Tn^2)$ which is far more as compared to the K-means $O(knT)$, additionally, the selection of the optimum values for various parameters in MS algorithm is non-adaptive.

Yu *et al.* [6] proposed ant-colony fuzzy c-means hybrid algorithm (AFHA), an adaptive unsupervised algorithm, which takes advantage of both ant colony system and FCM techniques. In [6], the ant system [7] determines the initialisation parameters while FCM clusters pixels to produce the final segmented image. The high-computational complexity of the ant system (AS) module decreases the overall efficiency of AFHA. To increase the overall efficiency of AFHA, Yu *et al.* proposed a modified version named IAFHA in [6]. IAFHA considered only a small portion of pixels instead of the whole image to determine the initialisation parameters. Even though IAFHA reduces the computational complexity yet obviously there is a trade-off between computational complexity and the algorithm performance which

affects the segmentation quality. In [8], Bhoyar and Kakde presented agglomerative just noticeable difference histogram (AJNDH) algorithm, which adopts histogram bins to determine the initial clusters. The number of histogram bins can be further reduced by merging perceptually similar colour segments to become larger segments. The decrement caused in the number of unique colours by the AJNDH makes it feasible for real-time applications. In [9], Chen *et al.* proposed a technique, which first converts an image to a multi-colour space followed by a peak finding algorithm to determine the initialisation parameters. Spatial FCM is applied after that to obtain initial segmentation results utilising the pre-determined number of clusters for each colour channel. In the final step, initial segmentation results are re-fused to obtain the final number of clusters. In [10], Zheng *et al.* proposed an adaptive image segmentation approach based on FCM and spatial information, the huge processing time of this approach makes it unsuitable for the real-time image segmentation.

In [11], Siang Tan and Mat Isa presented a hybrid approach for colour image segmentation based on the histogram thresholding and FCM clustering (HTFCM). The complexity in the initialisation parameter determination process decreased the segmentation quality and increased the convergence time of the clustering algorithm. In [12], Tan *et al.* presented an unsupervised adaptive clustering algorithm named region splitting and merging FCM hybrid algorithm (RFHA). RFHA consists of region splitting and merging (RSM), which determines the initialisation parameters followed by the production of the final segmented image by the FCM algorithm. RFHA is comparatively faster as compared to AFHA. In [13], Tan *et al.* proposed another image segmentation algorithm employing the hue-saturation-lightness (HSL) colour scheme instead of red–green–blue (RGB) colour channels. The final segmented image resulted in the production of relatively more homogeneous regions yet the computational complexity remained an issue. However, the deficiencies in the selection of initialisation parameters in HTFCM, RFHA and HA caused an increment in convergence time of FCM, thus causing the overall computational complexity to increase [14–18].

The modified K-means (MKM) algorithm presented by Zhang *et al.* [19] is another image segmentation based on adaptive unsupervised clustering. MKM repeats bisecting the clusters until

the inter cluster similarity is not reached. Thus, the value of the sensitivity threshold affects the performance of the MKM algorithm. In [20], Khan *et al.* presented an adaptive unsupervised algorithm for grey image segmentation using K-means clustering. The algorithm efficiently determined the initialisation parameters and segmented the grey image while preserving the information and object boundaries in the image.

In this study, an efficient clustering algorithm (ECA) for adaptive unsupervised colour image segmentation is presented that consists of a novel region splitting (RS) module and K-means clustering. In the RS module, firstly, peaks and the corresponding colour channel intensity levels are determined for all the colour channels using the histograms. Secondly, the individual colour channel intensity levels corresponding to the peaks are combined in an efficient way to construct the sets of RGB triplets. After that, perceptually close RGB triplets are merged into a single RGB triplet, decreasing the number of clusters to mitigate the computational complexity and the over-segmentation effect. The set of RGB triplets serve as the initialisation parameters while the amount of RGB triplets termed as the number of clusters. In the next step, K-means clusters the pixels in the image using the above-mentioned initialisation parameters.

The rest of the paper is organised as follows. Section 2 presents details of the RS module of the proposed approach and provides implementation steps and procedure of the ECA algorithm. Section 3 analyses the experimental results obtained from ECA algorithm and provides a qualitative and quantitative comparison with the existing approaches. Section 4 concludes the research work and results presented in this study.

2 Proposed approach

K-means being a local search method possess a tendency of falling into local minima during the iterative optimisation process. However, the performance of the K-means clustering technique is extremely sensitive to the initialisation parameters, i.e. initial cluster centres and the number of clusters. Thus, the image segmentation results produced by K-means are heavily dependent upon the selection of initialisation parameters. Usually, determining good initialisation conditions becomes a laborious process due to the need for experiments being performed.

In this study, an adaptive unsupervised ECA is proposed, ECA consists of RS module and K-means clustering. The proposed method ECA aims to improve the conventional K-means algorithm's colour image segmentation quality by optimising the determination of initialisation parameters. A block diagram is shown in Fig. 1, comprehensively presenting a general idea of the proposed ECA. The RS module is discussed in detail in Section 2.1.

In this study, a novel RS module is utilised as an optimised initialisation scheme determining the number of clusters and cluster centroids in an adaptive and non-parametric way. As compared to the commonly used random initialisation technique, the initialisation scheme based on the RS module is more robust and consistently provides significant initialisation parameters. The RS module detects a number of clusters and cluster centroids based on both the global information and the local information of an input image histogram. More specifically, every pixel in an RGB colour image is a triplet composed of red, green, and blue colour channel intensities. The combination of these three colour channels provides global information about an image. In the RS module, peak finding algorithm (PFA) is applied to determine the initialisation parameters for the K-means algorithm in a novel way.

First, PFA is applied on R, G, and B colour channels, determining significant peaks in each of the colour channel histograms. As each pixel in a RGB image is represented by a 3D feature vector, the two unknown channel intensity values are determined by applying PFA upon the histogram of the pixels corresponding to the initially determined peaks channel intensities. With the addition of two more colour channel intensity values in each of the initially determined peak channel intensity, RGB triplets are completed. In the next step, perceptually close RGB triplets are merged, producing the initialisation parameters for the K-means clustering algorithm. Applying K-means to an image provided better initialisation parameters, cluster pixels into appropriate regions while preserving both the colour features and the boundaries of various regions in the image. The flow diagram of our proposed approach is presented in Fig. 2.

In the following section, the details of the proposed RS module in ECA are discussed in detail.

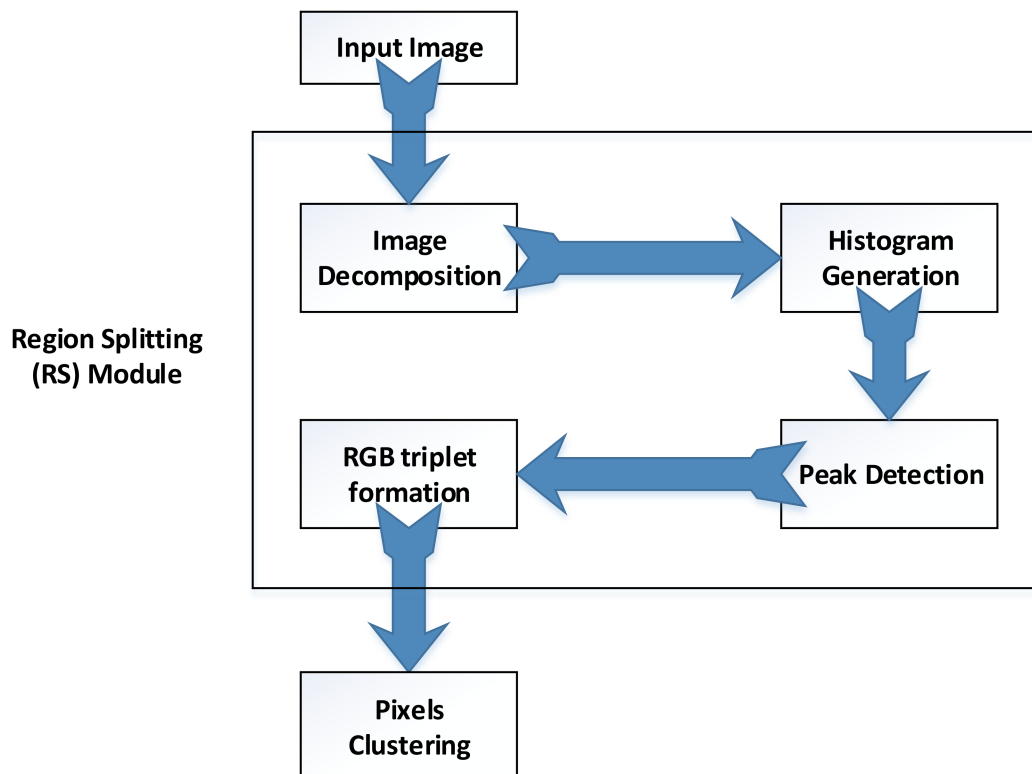


Fig. 1 Flow diagram of the proposed ECA algorithm

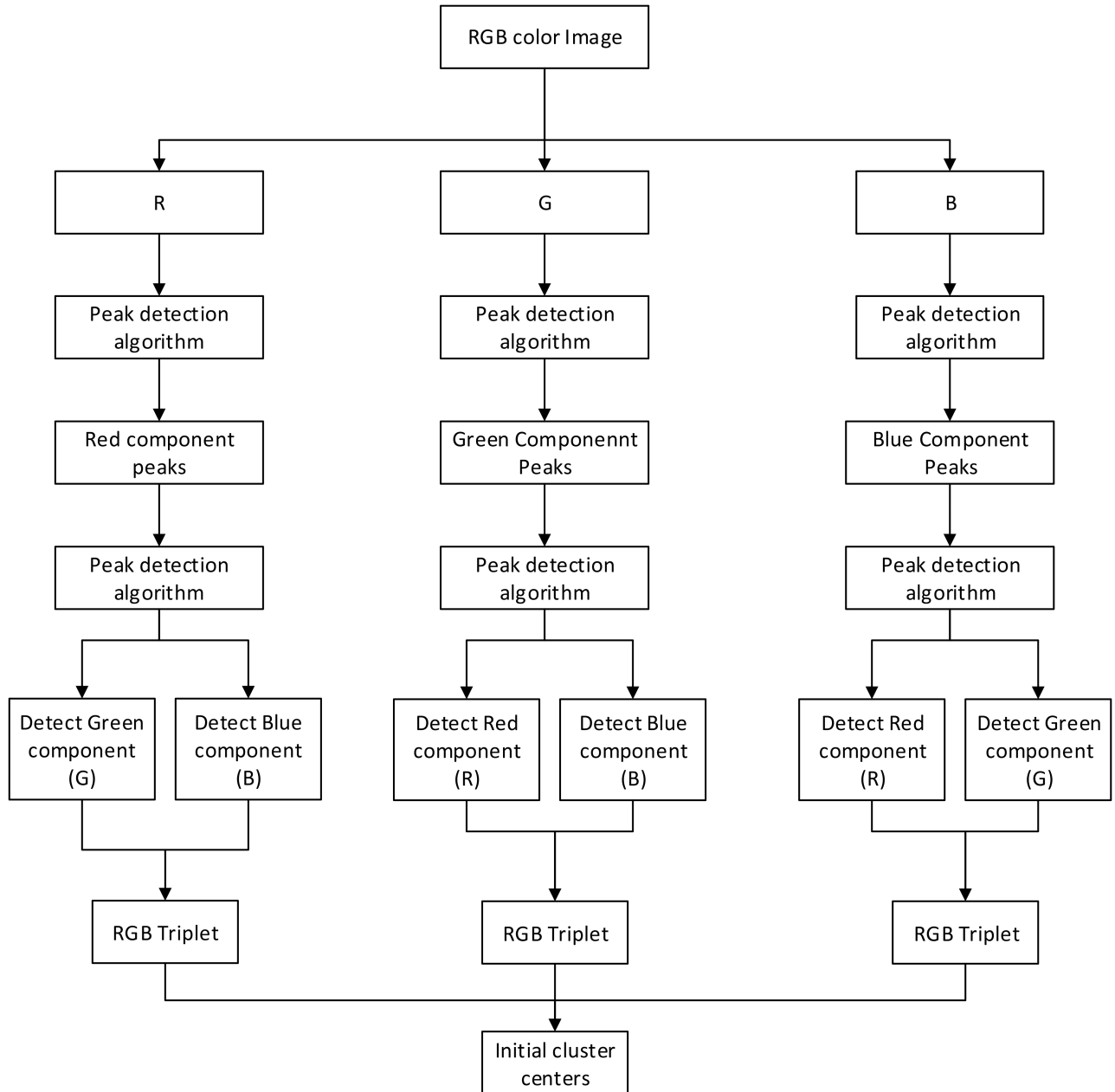


Fig. 2 Construction of RGB triplets

2.1 RS module

In this study, the RS module is proposed to enhance the efficiency of the K-means clustering algorithm for colour image segmentation. The RS module is proposed to determine the initialisation parameters for the K-mean clustering algorithm. The RS module is implemented as follows:

- Select significant peaks in each colour channel histogram by detecting the channel intensities with a maximum occurrence value among the neighbouring intensity levels. The peaks detected in the red, green, and blue colour channels are represented by an asterisk in Figs. 3a–c, respectively.
- Organise RGB triplets by determining the missing colour channel intensity levels by detecting the peaks in the histogram of the pixels belonging to the already detected colour channel intensity level. The peaks detected in the red colour channel are represented by an asterisk in Fig. 3b.
- Calculate the Manhattan distance between all the pairs of RGB triplets using the following equation. The detection of green and blue color channel peaks for a given red channel peak Fig. 3c are presented in Figs. 3(e) and 3(f) respectively:

$$D(c_i, c_j) = |R_i - R_j| + |G_i - G_j| + |B_i - B_j|, \quad \forall i \neq j, \quad (1)$$

where $1 \leq i \leq M$, $1 \leq j \leq N$, M and N represent the number of RGB triplets, R_i , G_i and B_i are the intensities of the red, green, and blue colour channels of the i th RGB triplet.

(Note: the Manhattan distance is found to be a better similarity measure than the Euclidean distance as Manhattan has a stable visual colour similarity whereas the later produces a wider variation of the same colour.) The segmentation results obtained after K-means clustering can be represented as $\bigcup_{i=1}^n c_i = C$, $\bigcup_{i=1}^n l_i = L$, and $\bigcup_{i=1}^n u_i = U$, where each cluster c_i is a set of pixels: $c_i = \{p_k\} 1 \leq k \leq N_i$, N_i is the number of pixels that belong to the cluster c_i , l_i is the label assigned to c_i cluster elements, u_i is the cluster centre of c_i and U is the set containing all the cluster centres. $\sum_{i=1}^n N_i = N$, where n is the number of clusters and N is the total number of pixels in the input image C (see Fig. 4).

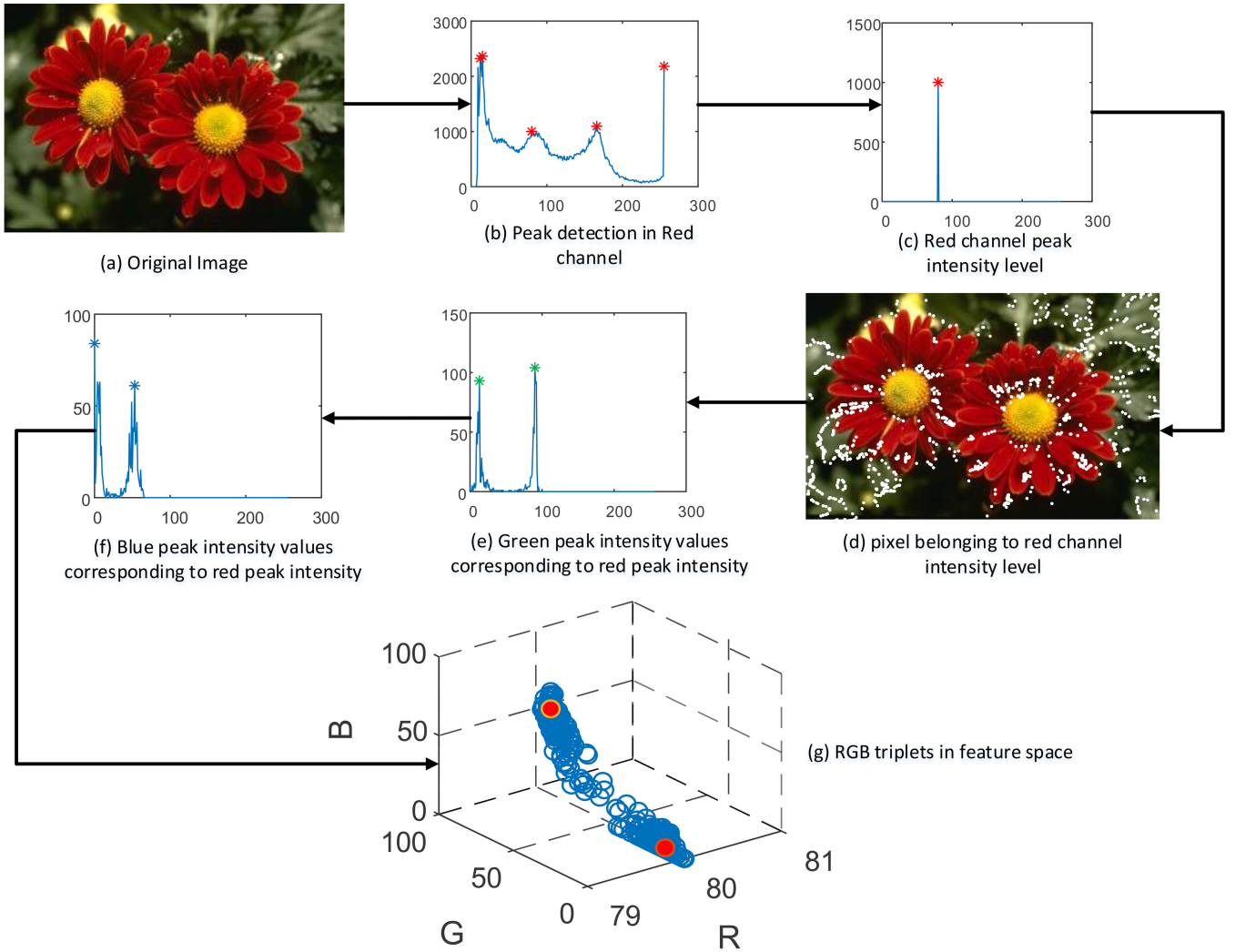


Fig. 3 Completion of RGB triplets for red channel intensity = 80

(a) Original image, (b) Peak detection in red channel, (c) Red channel peak intensity level, (d) Pixel belonging to red channel intensity level, (e) Green peak intensity values corresponding to red peak intensity, (f) Blue peak intensity values corresponding to red peak intensity, (g) RGB triplets in feature space

```

1   $i \leftarrow 1$ 
2   $RGBTriplet = \emptyset$ 
3  while  $i \leq 3$  do
4       $\{peaks\}_i = pfa(X_{p_i})$ 
5      for  $j \in \{peaks_i\}$  do
6           $\{pixels\}_{i,j} = C(\{peaks\}_{i,j})$ 
7           $\{peaks\}_t = pfa(\{pixels\}_{i,j})$ 
8          where  $t \in \{t_1, t_2\}$  and  $t_1, t_2 \neq i$ 
9          for  $m \in \{peaks_{t_1}\}$  do
10             for  $n \in \{peaks_{t_2}\}$  do
11                  $RGBTriplet(j, m, n)_i =$ 
12                      $\{(peaks)_i(peaks)_{t_1}(peaks)_{t_2}\}$ 
13             end
14         end
15     end
16 end
17  $RGBTriplet = \{RGBTriplet; RGBTriplet(j, m, n)_i\};$ 
18 end

```

Fig. 4 Algorithm 1: RGB triplet generation algorithm

2.2 Analysis of the computational complexity

The computational complexity of the mean shift algorithm is $O(n^2di)$ [21]. In the case of HA, HTFCM, and RFHA algorithms, the RSM processes have the computational complexities of $O(n^r n^g n^b)$ and $O(idc^2)$. FCM used as the clustering algorithm have a computational complexity of $O(nc^2di)$, where n^r , n^g and n^b

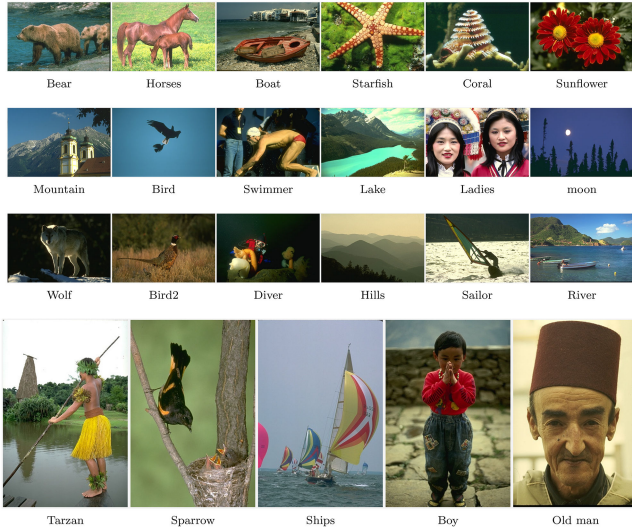
represent the peaks in each colour channel. Thus the cumulative computational complexity of the above-mentioned algorithms is $O(n^{r_0 n^{g_0} n^{b_0}}) + O(idc^2) + O(nc^2di)$ [22].

In the proposed algorithm, the application of the peak detection algorithm upon each individual colour channel has the computational complexity of $O(n^r) + O(n^g) + O(n^b)$, while the computational complexity of RGB triplets generation is $O(n^{g_1 \times b_1}) + O(n^{r_1 \times b_2}) + O(n^{r_2 \times g_2})$. Also, the computational complexity of K-means is $O(ncdi)$, least as compared to the rest of the clustering algorithms. Hence, the cumulative computational complexity of the proposed method becomes extremely low and is given as $O(n^{r \times g_1 \times b_1}) + O(n^{g \times r_1 \times b_1}) + O(n^{b \times r_1 \times g_1})$. It is worth mentioning that the computational complexity of RFHA's cluster splitting module has a multiplicative effect in the cluster centroids determination process, causing a surge in the algorithm's overall computational complexity. It is also evident that the RS module of the proposed ECA algorithm has fewer multiplicative terms, which leads to a significant reduction in the algorithm's overall computational complexity (Table 1).

n represent the number of data points, c represents the number of clusters, d represents the dimensions i is the number of iterations, r , g , and b represent the initial peaks detected by RFHA, HA, HTFCM, and the proposed algorithm while $g_1 \times b_1$, $r_1 \times b_2$ and $r_2 \times g_2$ represent the pairs of (green, blue), (red, blue), and (red, green) channel peak intensities detected by our proposed ECA algorithm.

Table 1 Comparison of computational complexity

Algorithm	Computational complexity
MS	$O(n^2 di)$
RFHA	$O(n^r n^g n^b) + O(idc^2) + O(nc^2 di)$
HA	$O(n^r n^g n^b) + O(idc^2) + O(nc^2 di)$
HTFCM	$O(n^r n^g n^b) + O(idc^2) + O(nc^2 di)$
ours	$O(n^{r \times g_1 \times b_1}) + O(n^{g \times r_1 \times b_1}) + O(n^{b \times r_1 \times g_1}) + O(nc di)$

**Fig. 5** Sample images from BSD500 dataset

3 Experimental results and analysis

The proposed algorithm was compared with the MATLAB implementation results of MS, RFHA, HA, and HTFCM algorithms. The latest version of the Berkeley segmentation data set and benchmark (BSD500) [21, 23] was selected to perform the experiments. Some sample images from BSD500 data set given in Fig. 5 are used to present the segmentation results given in Figs. 6–8. The first row for each sample image in Figs. 6–8 presents the segmented image while the second row shows the false colour representation of the segmented image clearly depicting the pixels membership to the clusters.

3.1 Qualitative analysis of the image segmentation results

By analysing Figs. 6–8, it becomes evident that the proposed algorithm (ECA) clearly outperforms other algorithms in terms of preserving original image features in the segmented image. For instance, MS, PHFA, RFHA, and HTFCM misclassified pixels on the body of bear, horses, boat, starfish and background while the proposed ECA preserved the pixel's features integrity by clustering the pixels to unique clusters having minute dissimilarity.

In case of the coral image, MS and RFHA resulted in an under-segmented image while HA and HTFCM over segmented the image, causing inhomogeneity in the background while the proposed ECA algorithm maintained the correct balance by segmenting the background as a homogeneous region and also preserved the coral foreground features. RFHA, HA, and HTFCM misclassified the pixels of the foreground regions in the sunflower image. RFHA, HA, and HTFCM misclassified pixels on the roof, tower top, and hills. In the case of a bird, MS over segmented the tail region, while in the case of swimmer image, MS falsely assigned the pixel on the personnel body to the background region causing misclassification error. In the case of bird and swimmer images, while RFHA, HA, and HTFCM misclassified pixels heavily, which can be observed by zooming into the figures. Pixels misclassification can be observed by RFHA, HA, and HTFCM in the river image segmentation. HTFCM performed false assignment of pixels in the ladies image.

For the Moon image, MS produced an under segmented image while the segmented images produced by RFHA, HA, and HTFCM resulted in over-segmentation causing classification errors while the proposed ECA maintains the trade-off between image details preservation and the production of homogeneous regions. It is evident that in the case of Tarzan image, segmented images produced by MS resulted in minor while RFHA, HA, and HTFCM resulted in major classification errors. MS under segmented the Sparrow image while RFHA, HA, and HTFCM misclassified the tree, background and the sparrow body pixels while observing the proposed ECA segmented image it is evident that correct pixels assignment to the clusters is accomplished. Under-segmentation and pixels misclassification can be observed in ships, boy and old man image by MS, RFHA, HA, and HTFCM, respectively.

3.2 Quantitative analysis of the image segmentation results

A quantitative comparison is performed to analyse the performance of clustered images produced by various algorithms using mean square error (MSE), F' value, and Q value. MSE is considered to be the most fundamental benchmark to assess cluster quality. MSE for the segmented image is calculated utilising the following equation:

$$MSE = \frac{1}{N} \sum_{i=1}^C \sum_{j \in p_i} \|x_j - c_i\|^2, \quad (2)$$

where N represents the total amount of pixels in the image, M is the number of clusters in the segmented image, p_i is the pixels set belonging to the i th cluster, x_j is the original feature vector of pixels in the i th cluster and c_i is the cluster centroid of the i th cluster. MSE actually measures the average difference between the original image pixels and the cluster centroids.

The results presented in Table 2 show that the proposed ECA algorithm produces the lowest MSE value for 94% of the sample images. RFHA only produced the lowest MSE for the Moon image. MS was unable to produce the lowest MSE for any of the sample images. After experimenting algorithms on 200 images, RFHA and HTFCM produced the lowest MSE for 3.5 and 6.5% of the images, respectively, while the proposed ECA algorithm validated its efficiency by obtaining the lowest MSE value for 86.5% of the images.

To further evaluate the performance of the proposed algorithm, few other famous benchmarks were utilised, measuring the segmented image quality by analysing the image regions. Borsotti *et al.* [24] first proposed F evaluation function, which penalises the over-segmented images by considering colour error and same area segments. The increment in either colour error or abundance of similar size segments causes an increase in the value of F . Borsotti further improved the benchmark by proposing another evaluation function, which penalises the tiny segments produced in the over-segmented images by considering the area of each segmented region in addition to the above-mentioned factors. The expressions to calculate $F'(I)$ and $Q(I)$ are given in (3) and (4), respectively

$$F'(I) = \frac{\sum_{j=1}^M e_j^2 \sqrt{\sum_{a=1}^{\max Area} [S(a)]^{1+(1/a)}}}{(1000 \times N) \sqrt{N_j}}, \quad (3)$$

$$Q(I) = \frac{1}{1000 \times N} \sqrt{M} \sum_{j=1}^M \left[\frac{e_j^2}{1 + \log N_j} + \left(\frac{S(N_j)}{N_j} \right)^2 \right], \quad (4)$$

where N represents the total number of pixels in image I , M represents the number of segments, N_j is the number of pixels in the j th segment, $S(a)$ represents the number of regions in image I having an area a and Max Area denotes the largest region in the segmented image. By analysing the results given in Table 3, it becomes evident that the proposed ECA attains the lowest $F'(I)$ and $Q(I)$ for a majority of sample images. RFHA attained the lowest $F'(I)$ and $Q(I)$ value for 1.5% sample images while HTFCM

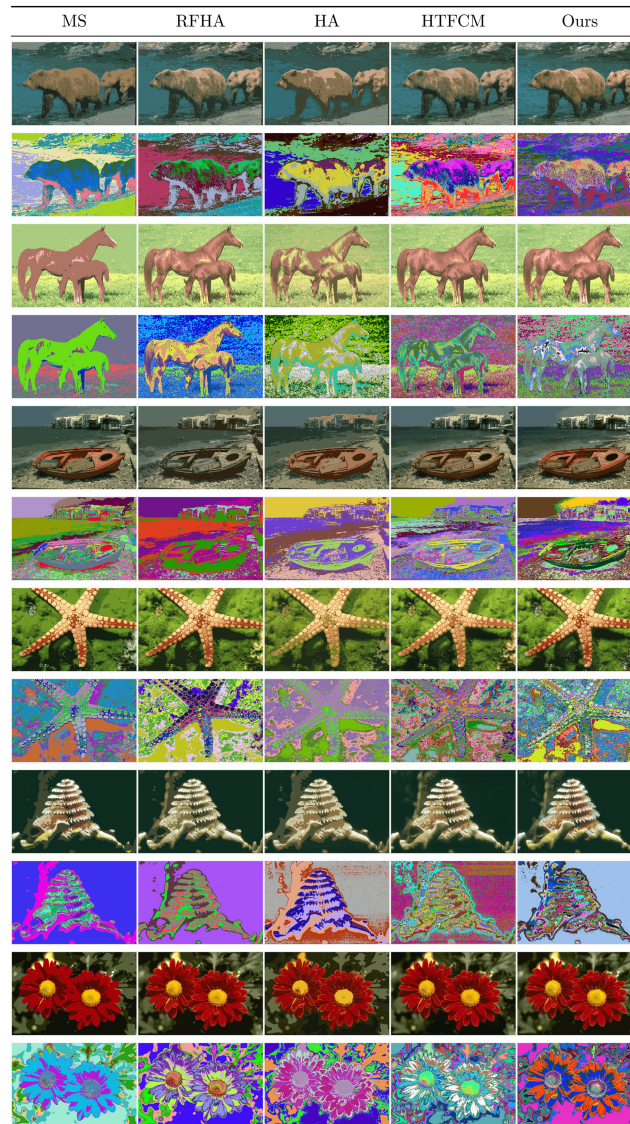


Fig. 6 Sample images from BSD500 dataset

for 31 and 45%, respectively. The proposed ECA achieves the lowest $F'(I)$ and $Q(I)$ values for 82 and 77% of the sample images.

3.3 Execution time analysis

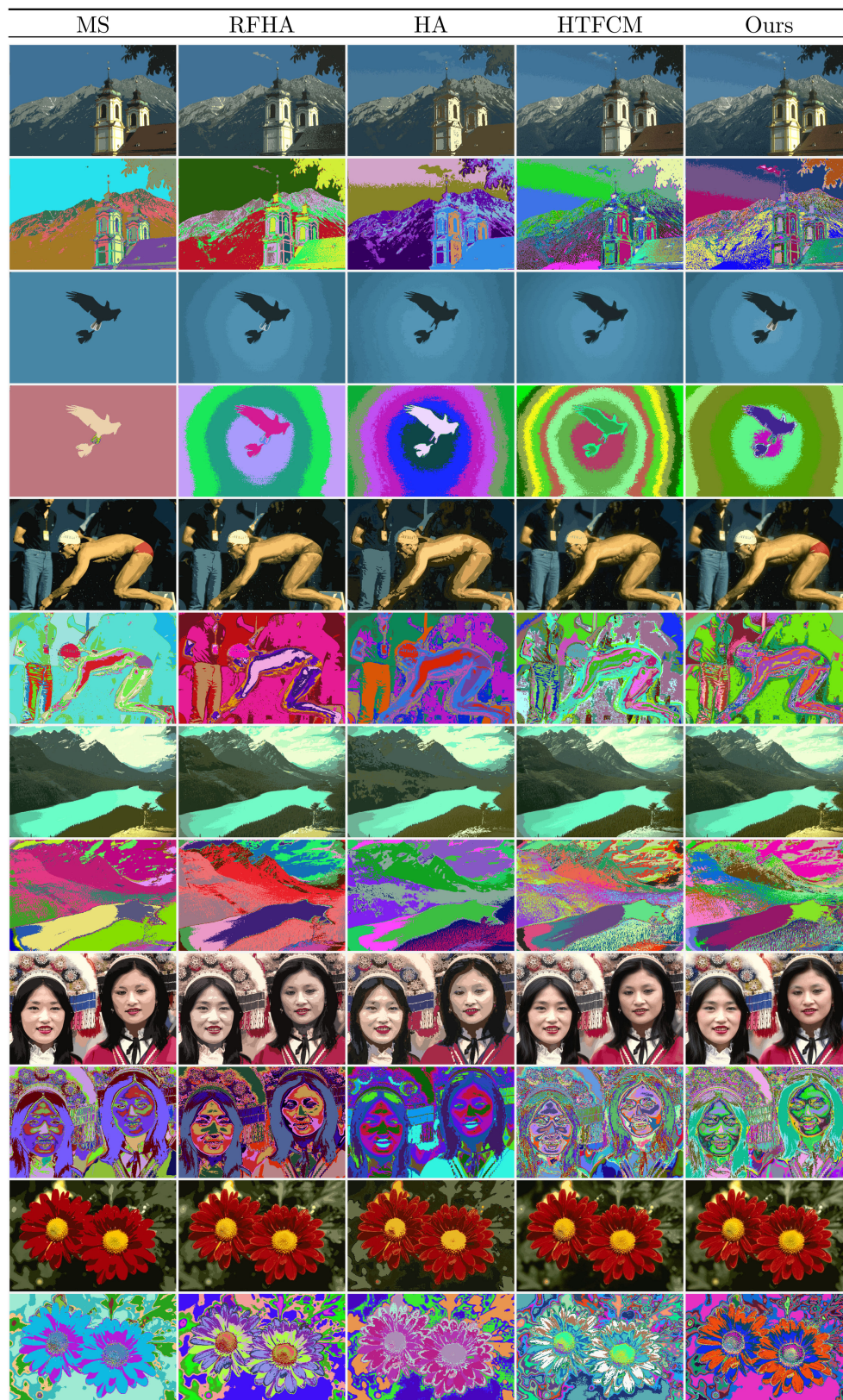
Several experiments were conducted to compare the execution speed of the proposed ECA algorithm with MS, RFHA, HA, and HTFCM. Table 4 shows the execution speed of each algorithm for ten sample images processed in MATLAB 2017b using an Intel Core i5-7400@3.00 GHz processor. Also, the execution time for 200 images from BSD500 data set is reported in Table 5. Analysing the results provided in Tables 4 and 5, we conclude that the proposed algorithm also outperforms MS, RFHA, HA, and HTFCM in terms of computational complexity. For the proposed algorithm, the average execution time is 1.7 s, much lower than MS, RFHA, HA, and HTFCM. Average execution time per image for MS is calculated to be 68 s and for RFHA, HA, and HTFCM, the average execution time is 4, 7 and 8 s, respectively. The statistics provided in Table 5 clearly depict that the efficiency of the proposed ECA algorithm is far better in terms of cluster quality as well as the execution time. The extremely lower execution time of the proposed algorithm makes it really suitable for real-time image segmentation applications.

4 Conclusion

In this study, a novel adaptive unsupervised segmentation algorithm based on the histogram thresholding is proposed to generate high-quality segmented images. The proposed algorithm

is capable of automatic determination of the significant number of clusters and optimised cluster centroids for a given set of pixels. In this proposed method, firstly individual peak colour channel intensities are determined based on the gradient change, then in the next step, the intensity levels in each histogram are screened by selecting the highest peak intensity level in the neighbouring intensity levels range. These individual peaks are further explored by examining histogram of pixels belonging to the peak intensity levels and thus selecting the peak intensity levels to obtain a complete RGB triplet. Owing to the possible abundance and duplication of RGB triplets, a region merging step is introduced to merge the perceptually closer triplets. In the final step, the RGB triplets serve as the initialisation parameters for the clustering algorithm. The exploration of the channel peak intensities to complete the RGB triplet instead of permutation of the individual peaks in each channel histogram causes a tremendous reduction in the computational complexity of the proposed ECA algorithm.

Also, the experimental results reported in the study validate the image segmentation quality of the proposed algorithm in comparison with MS, RFHA, HA, and HTFCM based on various unsupervised image segmentation evaluation benchmarks. Moreover, the experimental results on a variety of images prove the proposed algorithm is extremely agile as compared to other algorithms reported in this study, making the proposed algorithm perfectly suitable for real-time image segmentation applications.



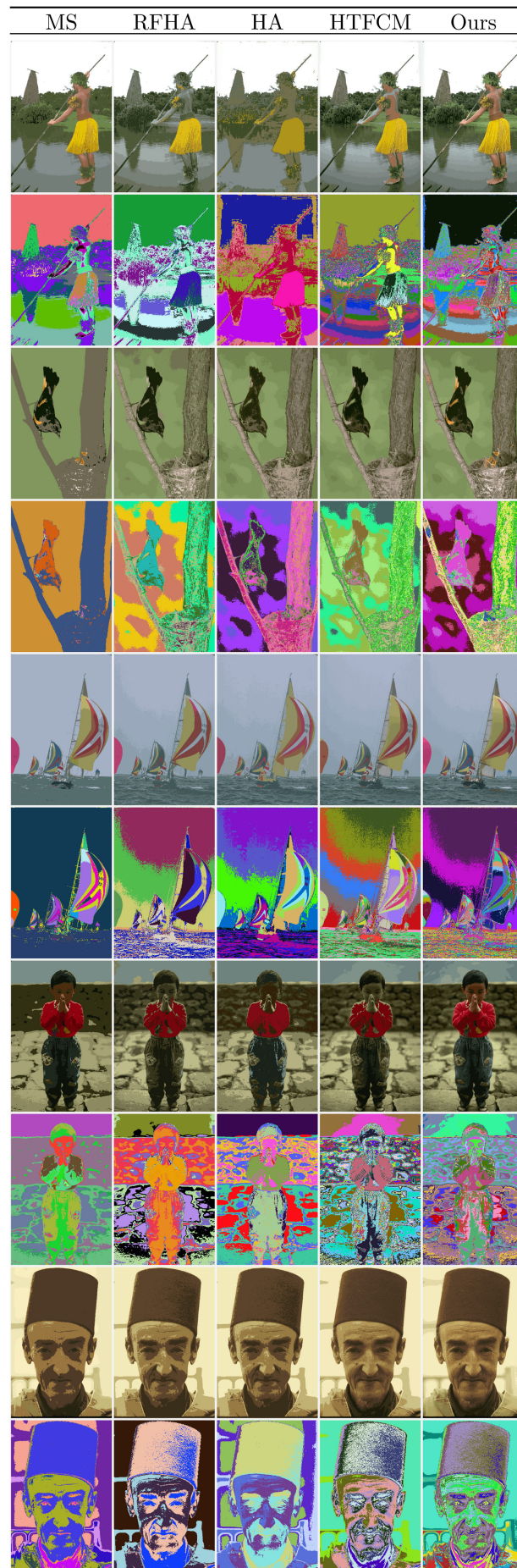


Fig. 8 Sample images from BSD500 dataset

Table 2 Comparison between the amount of detected initial cluster centroids and MSE for different algorithms

Image	No. of clusters					MSE				
	MS	RFHA	HA	HTFCM	Ours	MS	RFHA	HA	HTFCM	Ours
bear	24	8	7	10	34	0.2769	0.1615	0.2566	0.1294	0.0388
tarzan	76	9	7	16	79	0.2026	0.1697	0.5129	0.1096	0.0326
horses	34	9	7	3	26	0.4476	0.1963	0.2805	0.6270	0.0965
boat	30	5	6	15	44	0.1403	0.3619	0.4809	0.1364	0.0419
starfish	65	10	9	18	58	0.2265	0.1882	0.3362	0.1961	0.0518
coral	50	6	6	9	64	0.2475	0.1454	0.3281	0.1150	0.0286
sunflower	146	16	9	11	77	0.2274	0.0976	0.2486	0.1494	0.0434
mountain	61	8	8	14	57	0.1250	0.1545	0.2926	0.0859	0.0277
bird	29	6	7	10	26	0.1744	0.0230	0.0414	0.0192	0.0128
swimmer	77	10	11	15	70	0.1439	0.1392	0.3203	0.1339	0.0235
sparrow	65	8	9	24	34	0.5098	0.1015	0.1673	0.0572	0.0389
ships	54	9	10	27	65	0.1695	0.0849	0.1419	0.0454	0.0164
lake	45	10	9	13	65	0.2446	0.1713	0.5059	0.1713	0.0382
boy	67	8	8	21	36	0.3131	0.1684	0.2845	0.0839	0.0356
ladies	127	10	11	18	147	0.2387	0.2484	0.6312	0.1774	0.0302
old man	14	5	7	10	25	0.1784	0.1552	0.1832	0.0676	0.0340
moon	18	6	5	4	15	0.0639	0.0209	0.1031	0.1043	0.0315

Bold values indicate the best performing method for a given evaluation measure.

Table 3 Comparison of $F(I)$ and $Q(I)$ values of different algorithms after clustering

Image	$F(I)$					$Q(I)$				
	MS	RFHA	HA	HTFCM	Ours	MS	RFHA	HA	HTFCM	Ours
bear	0.0161	0.0060	0.0071	0.0044	0.0015	0.2061	0.0860	0.0974	0.0511	0.0125
tarzan	0.0222	0.0045	0.0117	0.0029	0.0015	0.2209	0.0556	0.1530	0.0272	0.0110
horses	0.0510	0.0057	0.0075	0.0199	0.0038	1.0210	0.0680	0.1013	0.3805	0.0381
boat	0.0071	0.0112	0.0135	0.0043	0.0016	0.0826	0.1784	0.1944	0.0414	0.0125
starfish	0.0204	0.0055	0.0080	0.0061	0.0019	0.2676	0.0705	0.0986	0.0598	0.0142
coral	0.0189	0.0039	0.0088	0.0031	0.0013	0.2886	0.0546	0.1208	0.0324	0.0150
sunflower	0.0301	0.0028	0.0069	0.0039	0.0029	0.4141	0.0298	0.0873	0.0394	0.0339
mountain	0.0130	0.0054	0.0066	0.0023	0.0015	0.1984	0.0792	0.0832	0.0240	0.0158
bird	0.0304	0.0007	0.0003	0.0003	0.0011	0.9047	0.0110	0.0037	0.0030	0.0178
swimmer	0.0129	0.0041	0.0089	0.0032	0.0010	0.2073	0.0524	0.1079	0.0307	0.0090
sparrow	0.1021	0.0029	0.0024	0.0012	0.0020	1.6923	0.0395	0.0262	0.0099	0.0220
ships	0.0286	0.0020	0.0025	0.0011	0.0009	0.4514	0.0240	0.0263	0.0074	0.0106
lake	0.0151	0.0055	0.0136	0.0048	0.0017	0.2033	0.0705	0.1691	0.0494	0.0135
boy	0.0461	0.0052	0.0071	0.0021	0.0013	0.4613	0.0687	0.0888	0.0186	0.0106
ladies	0.0392	0.0074	0.0175	0.0052	0.0012	0.3348	0.0860	0.2043	0.0471	0.0069
old man	0.0099	0.0052	0.0045	0.0028	0.0019	0.1710	0.0828	0.0654	0.0395	0.0251
moon	0.0043	0.0005	0.0009	0.0008	0.0023	0.0917	0.0070	0.0153	0.0150	0.0525

Bold values indicate the best performing method for a given evaluation measure.

Table 4 Comparison of execution times of different algorithms for sample images

Image	Image size	Execution time, (sec)				
		MS	RFHA	HA	HTFCM	Proposed
bear	481 × 321	75.5905	4.7444	6.0541	5.5077	2.2703
tarzan	481 × 321	58.9073	6.2269	8.2906	8.1756	3.3450
horses	481 × 321	182.8920	5.0291	4.8826	1.9379	2.0159
boat	481 × 321	41.4715	4.1693	5.6405	10.3598	2.5894
starfish	481 × 321	161.9011	6.0418	5.8447	11.8189	2.6225
coral	481 × 321	66.3923	4.7587	5.9636	5.9188	2.7038
sunflower	481 × 321	55.9233	7.6771	7.2326	5.0873	3.2098
mountain	481 × 321	24.5990	5.5137	6.1534	6.9641	2.5667
bird	481 × 321	10.4153	5.3294	11.9390	4.3410	1.8598
swimmer	481 × 321	33.1727	6.4501	12.2732	7.1841	2.8153
sparrow	481 × 321	108.5159	4.6803	5.5826	10.5815	2.0545
ships	481 × 321	37.9697	5.3398	7.1970	11.4602	2.6691
lake	481 × 321	55.0997	6.0747	7.4768	6.3033	2.7066
boy	481 × 321	53.7051	4.9681	6.2004	8.8684	2.0387
ladies	481 × 321	96.5289	6.3453	19.9213	8.6477	5.0196
old man	481 × 321	20.9642	4.9018	8.4798	5.1169	1.7979
moon	481 × 321	3.6252	7.6538	8.3664	2.6821	1.6546

Bold values indicate the best performing method for a given evaluation measure.

Table 5 Comparison of experimental results of different algorithms for 200 images from BSD500 data set

Evaluation function	MS	RFHA	HA	HTFCM	Ours
best MSE	3.5%	3.5%	0%	6.5%	86.5%
best $F'(I)$	0.50%	1.5%	0.5%	31%	66.5%
best $Q(I)$	0%	1.5%	0.5%	45%	53%
best execution time	0.50%	0%	0%	0%	99.50%
average execution time, s	67.5946	4.5227	7.7894	8.1511	1.7031

Bold values indicate the overall efficiency of various compared methods.

6 References

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